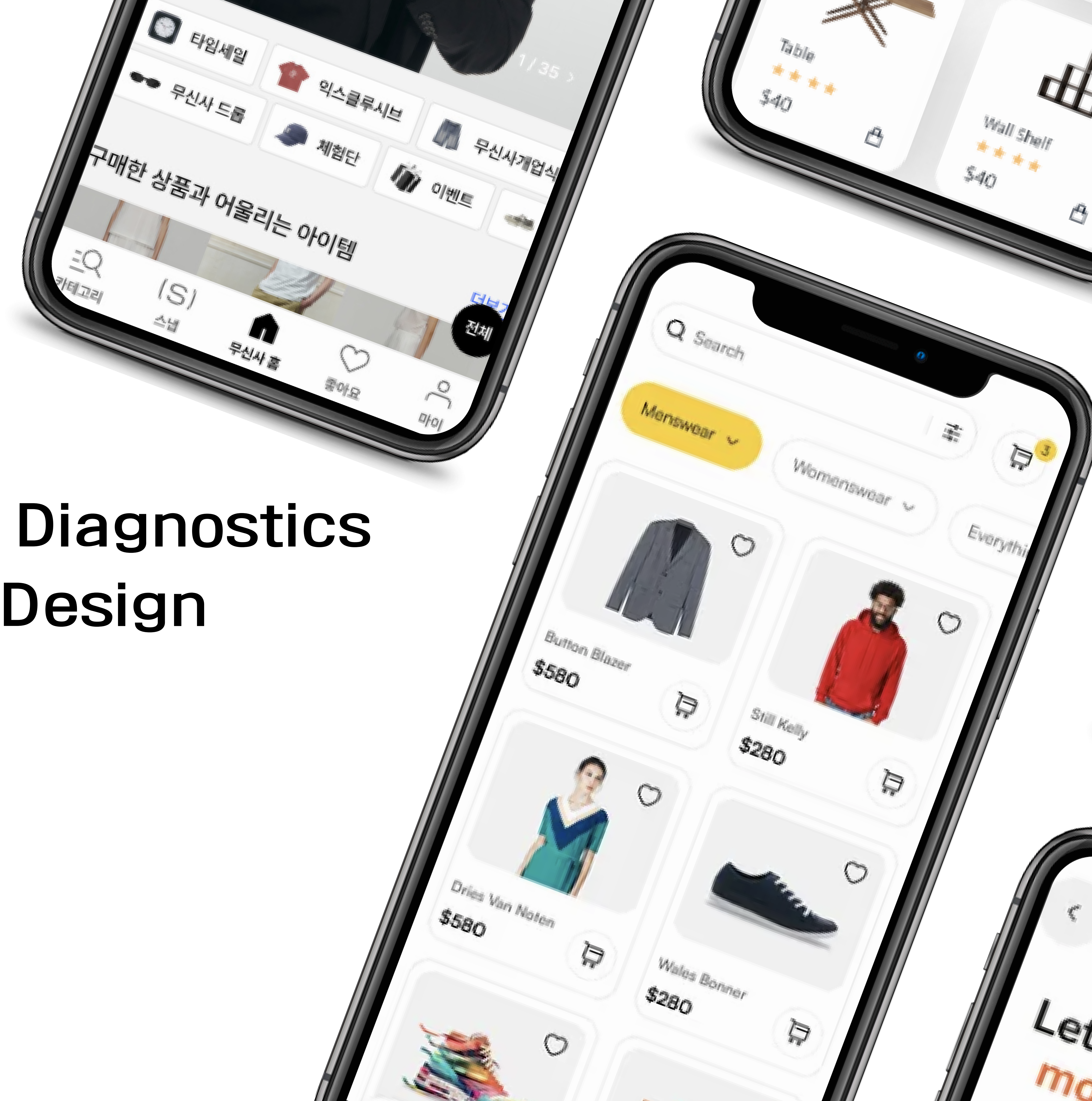


XAI-Based UI Complexity Diagnostics for Better Shopping App Design

Team 6 | Public Portfolio Copy

Member names and student IDs removed

Content

Goal of Analysis

Dominant Share: Mobile transactions account for **77.8%** of total domestic shopping volume.

High Adoption Rate: Over **91.8%** of e-commerce users have experienced or actively use mobile shopping.

2025년 6월 온라인쇼핑동향 및 2/4분기 온라인 해외 직접 판매 및 구매 동향

담당자 **윤주영** | 담당부서 **서비스업동향과** | 전화번호 **042-481-2195** | 게시일 **2025-08-01** | 조회 **31907**

2일 국가데이터처가 발표한 '2025년 11월 온라인쇼핑동향'을 보면 작년 11월 온라인쇼핑 거래액은 24조 1613억원으로 집계됐다. 전년 같은 달에 비하면 1조 5306억원(6.8%) 늘었고, 전달에 비해서도 1조 2982억원(5.7%) 증가하면서 역대 최고치를 기록했다.

온라인쇼핑 중 모바일쇼핑 거래액은 18조 5941억원으로 7.9% 늘었다. 전체 온라인쇼핑 거래액 중 모바일쇼핑 거래액 비중은 77.0%로 전년동월(76.1%)에 비해 0.9%포인트 높아졌다.

AI 구독하는 소비자 늘었다...소비자 10명 중 7명 디지털 소비

이다온 기자 · 2026. 5. 31. 15:52

소비자의 73.1%가 전자상거래를 통해 소비한 것으로 나타났으며 이는 2019년 이후 지속적으로 증가했다. 거래 유형으로는 '모바일 쇼핑' 이용률이 91.8%로 가장 높았고, 금융 플랫폼 이용률은 45.3%로 2023년 대비 7.0%p 상승해 가장 큰 상승 폭을 보였다. 반면'라이브 커머스' 이용률은 9.4%로 가장 낮게 나타났다.

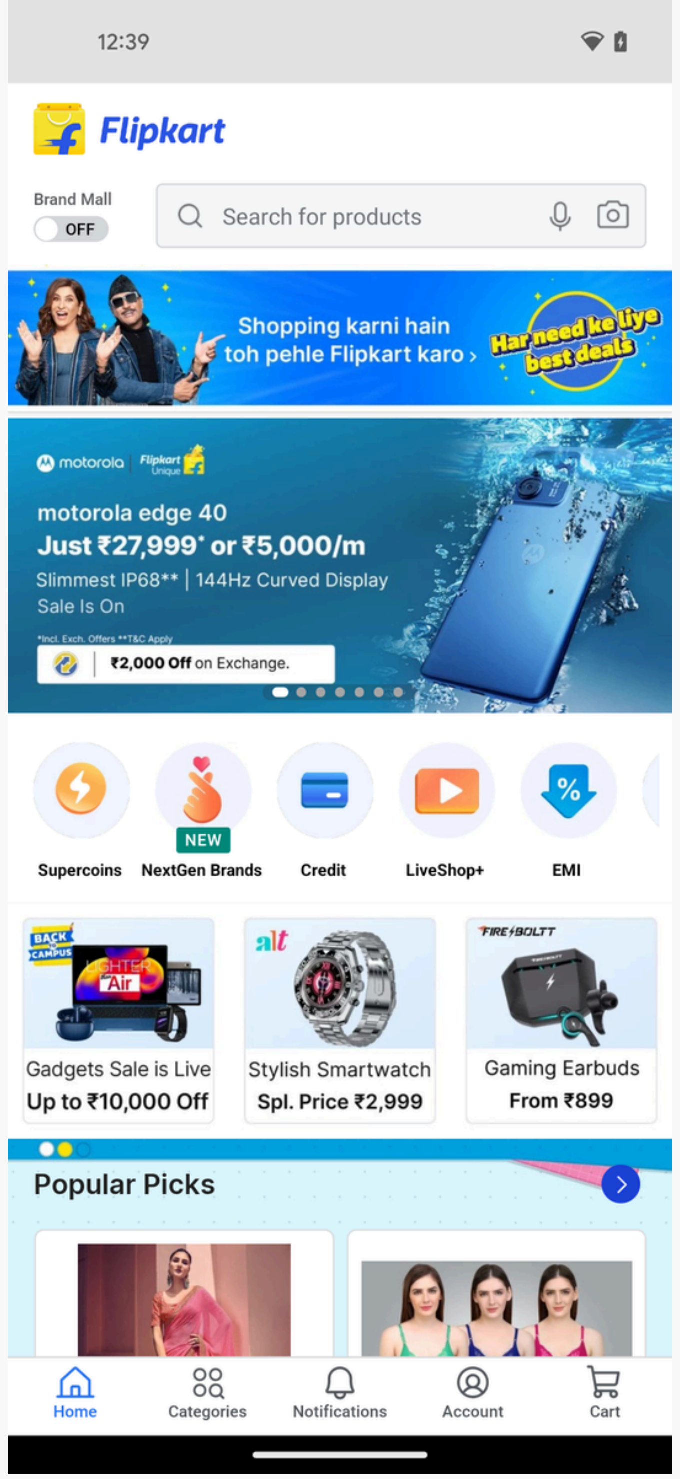
Goal of Analysis

Goal of Analysis

Expected Impact

- **The Golden Line:** Balancing Business Needs vs. User Experience.

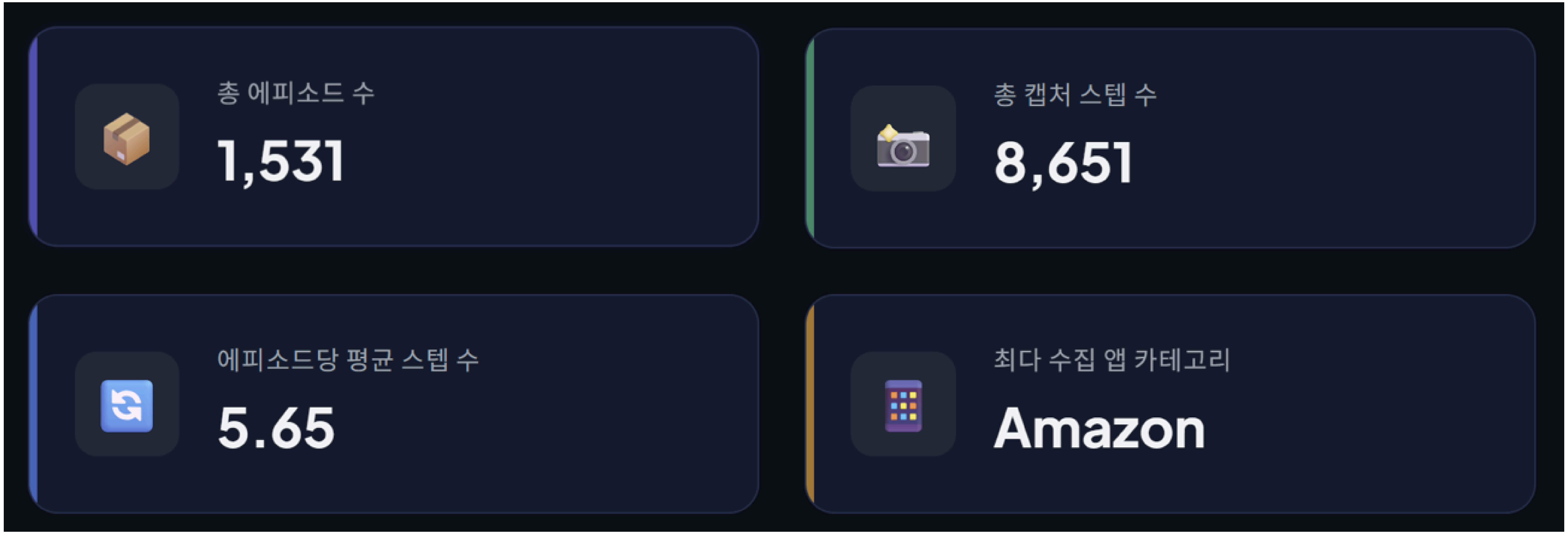
- **Model Optimization:** Maximizing Explainability + Lightweight Architecture.



Research Pipeline

1. **Baseline Metrics:** Parsing Google Research JSON
2. **Feature Expansion:** Advanced UI variables selection & Multicollinearity check.
3. **Final Model:** Quantitative evaluation for real-world UI usability.

Data



- Data Source: Google Research's Android Control Dataset (Public Open-Source)
- 1531 episodes(8651 Screenshots & JSON file)

Data

에피소드 인벤토리

에피소드 ID	앱 이름 (Task Type)	대표 패키지명	총 스텝 수	주요 화면 유형
# 5	Amazon	in.amazon.mShop.android.shopping	5 스텝	Product List
# 14	Ebay	com.ebay.mobile	7 스텝	Product List
# 16	Ticktick	com.ticktick.task	5 스텝	Product List
# 30	Etsy	com.etsy.android	4 스텝	Product Detail

Data

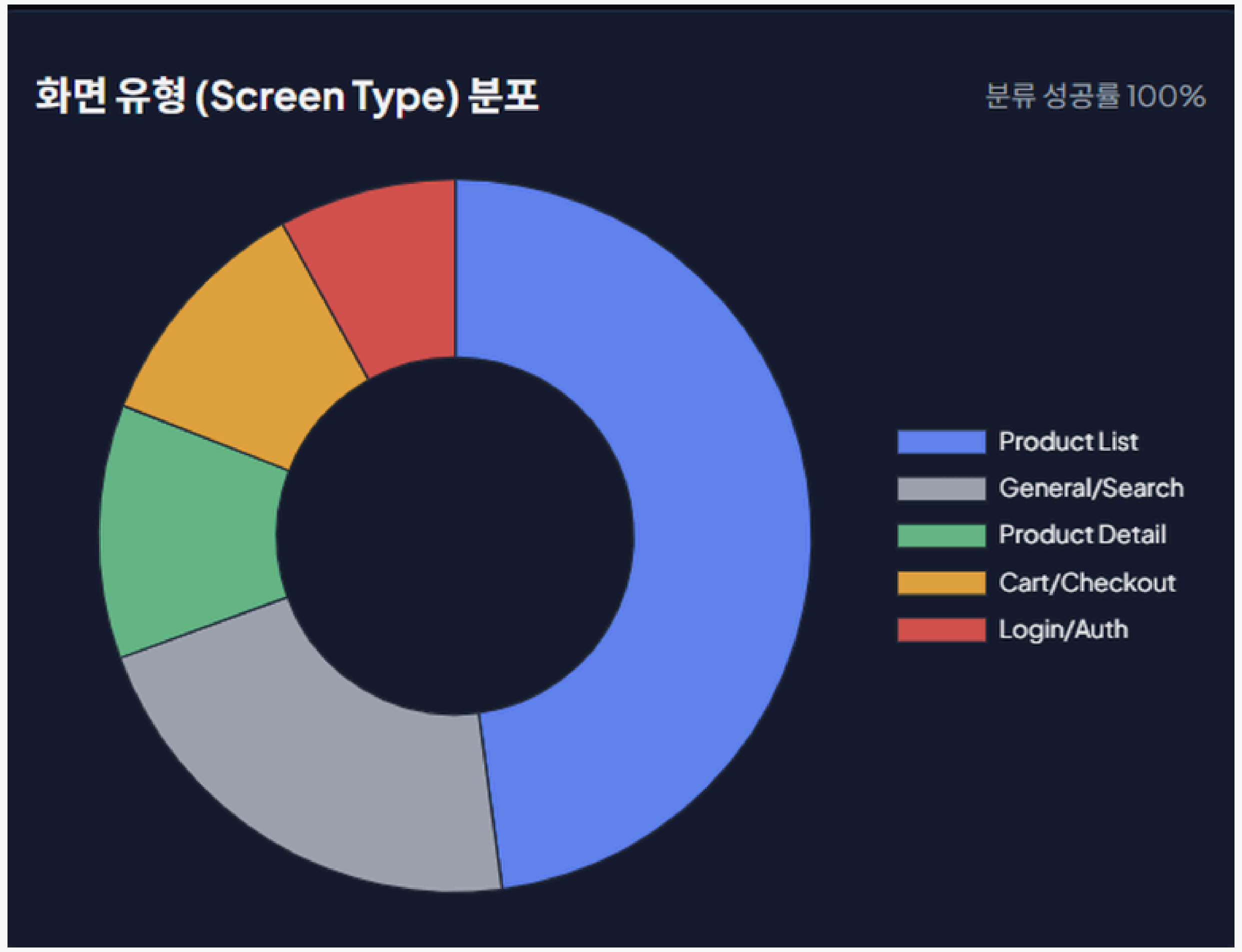
Number of X_feature : 45, (Ratio : 36, Interval Scale : 9)

Category	X_feature
Basic Structure (7)	num_elements, num_text_elements, num_clickable, etc.
Text (6)	text_density, total_text_length, mean_text_length, etc.
Interaction (5)	clickable_density, scrollable_density, input_density, etc.
Structural Hierarchy (5)	depth_mean, depth_max, depth_std, etc.
Layout Density (5)	area_density, whitespace_ratio, local_density_mean, etc.
Layout Alignment (4)	layout_alignment_score, grid_quality_score, etc.
Balance (5)	balance_score, equilibrium_score, etc.
Shopping Semantics (8)	price_text_count, discount_text_count, etc.

Data

Number of Domain_feature : 3

Doamin Feature	Example
Step_id (1531)	episode_5_step_0
app_name	Amazon, Flipkart, Adidas, Unknown, etc
screen_type (5)	Product List, General/Search, Cart/Checkout, etc



Target y, (Derivation Methodology)

$$Y_{target} = 9.85 + 0.39X_1 + 0.29X_2 + 0.23X_3$$

After standardizing the extracted variables, we perform Factor analysis using the academic standard Bartlett's Score method.

X_1

Dominant Colors:

- The number of perceptually salient color clusters across the screen.

X_2

Visual Clutter:

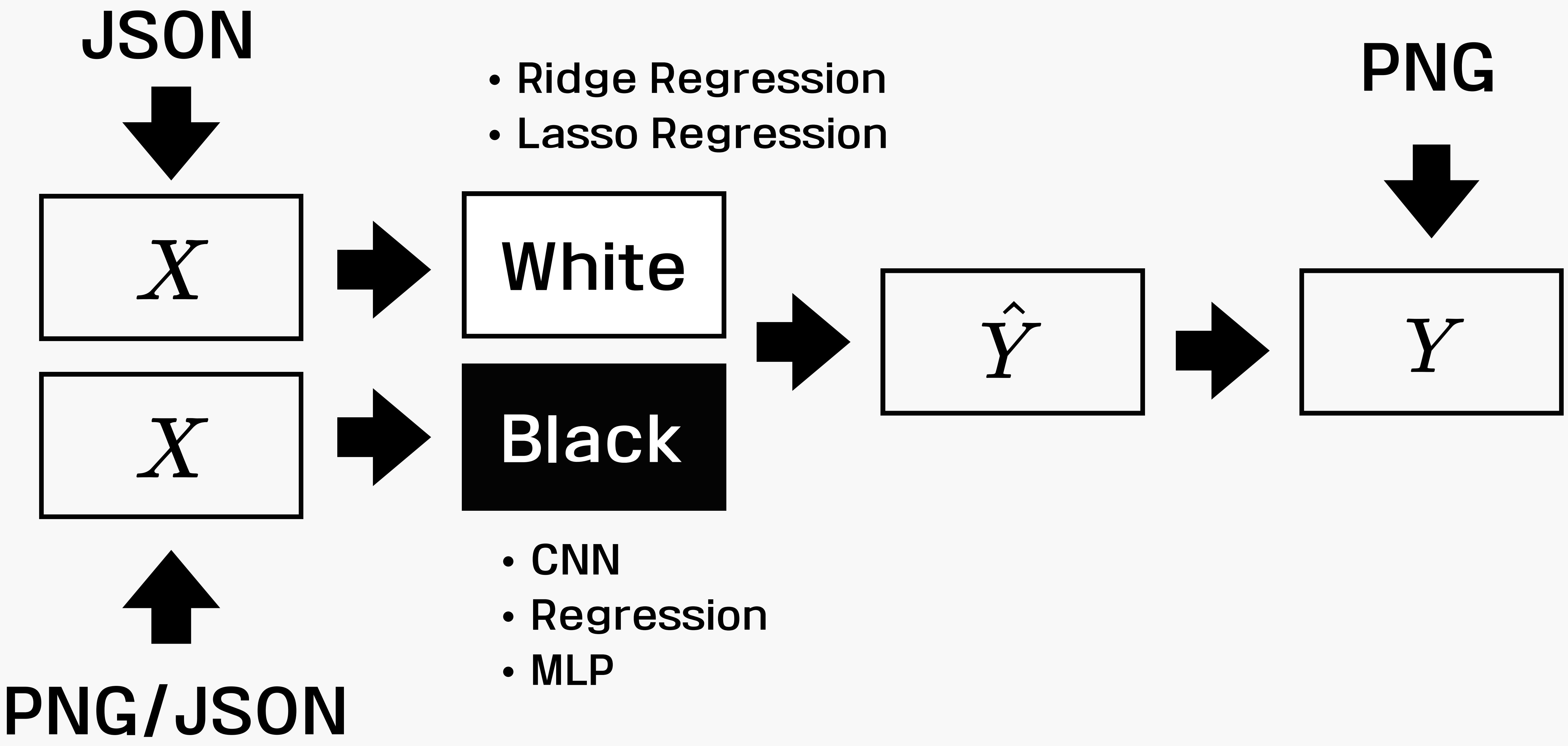
- The overall visual information load and spatial disorder of the UI.
- It also includes sub-variables.

X_3

Edge Congestion:

- The localized crowding of visual boundaries within the layout.

Analysis Techniques



WHITE BOX Model

Ridge Model

Analysis Techniques

Phase 1 : Filter

Metadata Integration

- Categorization by Screen Type

Exclusion of Low-Performance Domains

- Remove of Login/Auth , Product Detail type

Removal of Background & Home Screens

- Step_inx == 0 and num_UI < 50

Filtering of Non-Functional Blank Screens

- X_num_text_elements < 2
- Black and White Screen

Phase 2 : Rescale

Outlier Trimming on Target Variable (Y)

- Removal of top 1% and bottom 1% of the baseline y_target distribution

100-Point Readability Rescaling

- Linear transformation and inversion of C within the empirical range [8.0, 11.0]

Analysis Techniques

Phase 3 : Prune

Feature Pruning : Derived from a formula

- pruning of redundant variables

Remove Directly Redundant Features

- Dropping **avg_children_count,** overlap with **depth_mean** to prevent weight bias.

Sequential VIF Purification Algorithm

- Retaining only the core 28 features

Phase 4 : Data Isolate

Episode-based Split

- Grouped Train/Test split by episodes to prevent data leakage.

Skewness Correction

- Log transformation applied to highly skewed features.

Missing Value Handling

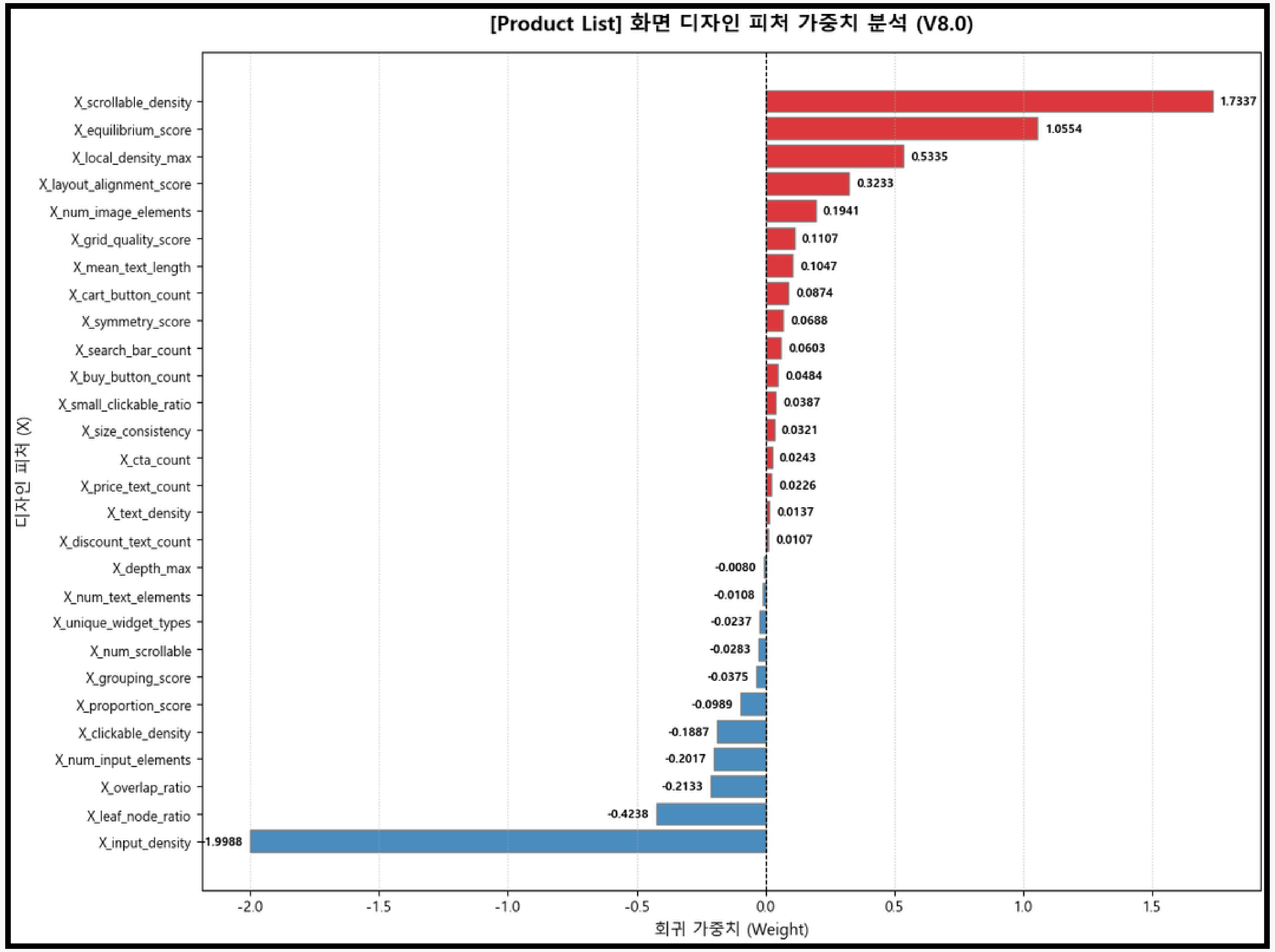
- Removed and imputed null data.

Analysis Techniques

	R^2	MSE
Product List	0.3216	0.2219
General/Search	0.3994	0.2267
Cart/Checkout	0.2078	0.1202

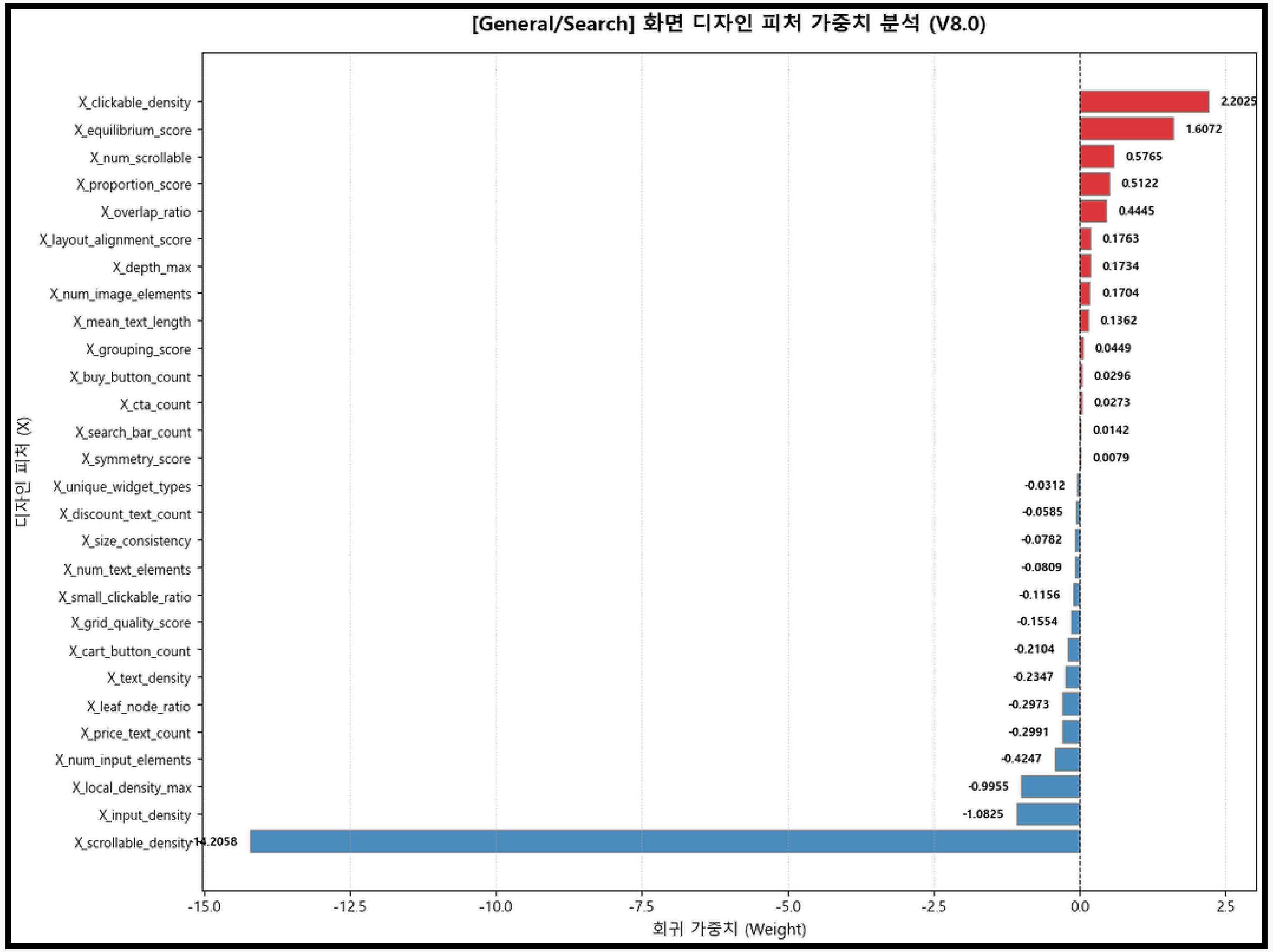
"A valid moderate R^2 (0.3-0.4) by HCI standards,
ensuring highly reliable and causal UI prescriptions."

Results



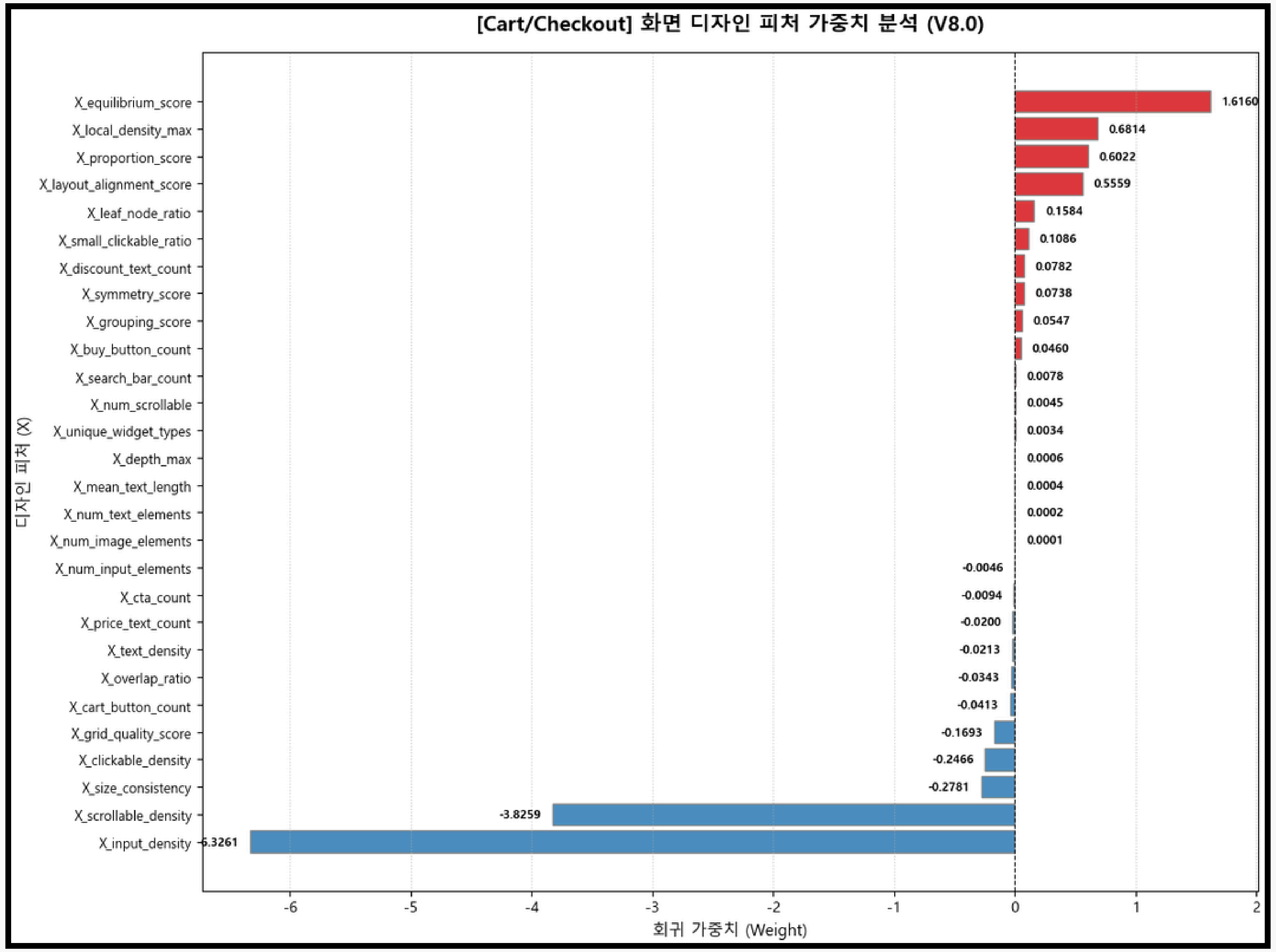
1. Product List
   - Top Complexity Inducers :
     - X_scrollable_density : +1.7337
     - X_equilibrium_score : +1.0554
     - X_local_density_max : +0.5335
   - Top Complexity Reducers :
     - X_input_density : -1.9988
     - X_leaf_node_ratio : -0.4238
     - X_overlap_ratio : -0.2133

Results



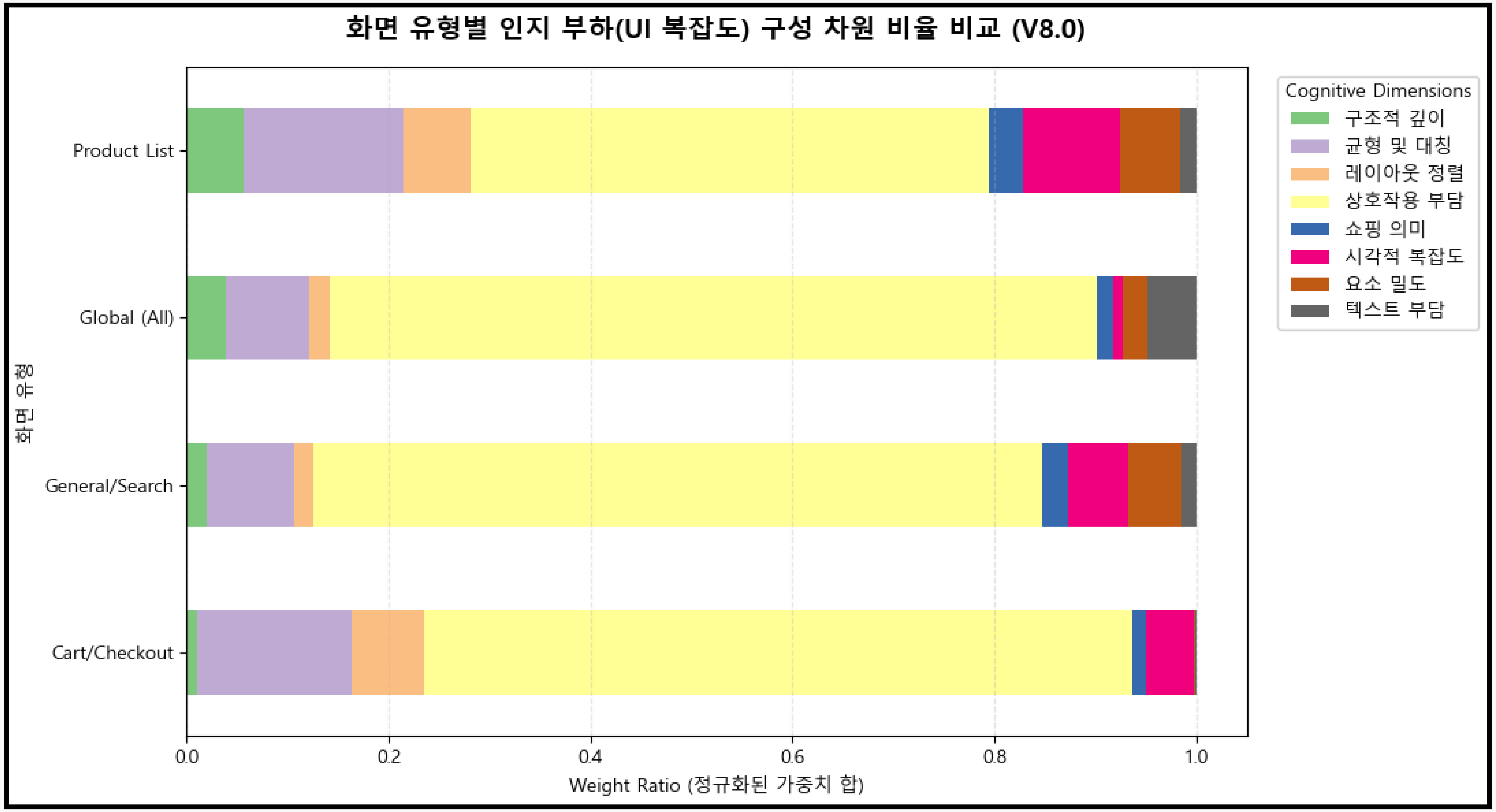
2. General/Search
- Top Complexity Inducers :
  - X_clickable_density : +2.2025
  - X_equilibrium_score : +1.6072
  - X_num_scrollable : +0.5765
- Top Complexity Reducers :
  - X_scrollable_density : -14.2058
  - X_input_density : -1.0825
  - X_local_density_max : -0.9955

Results

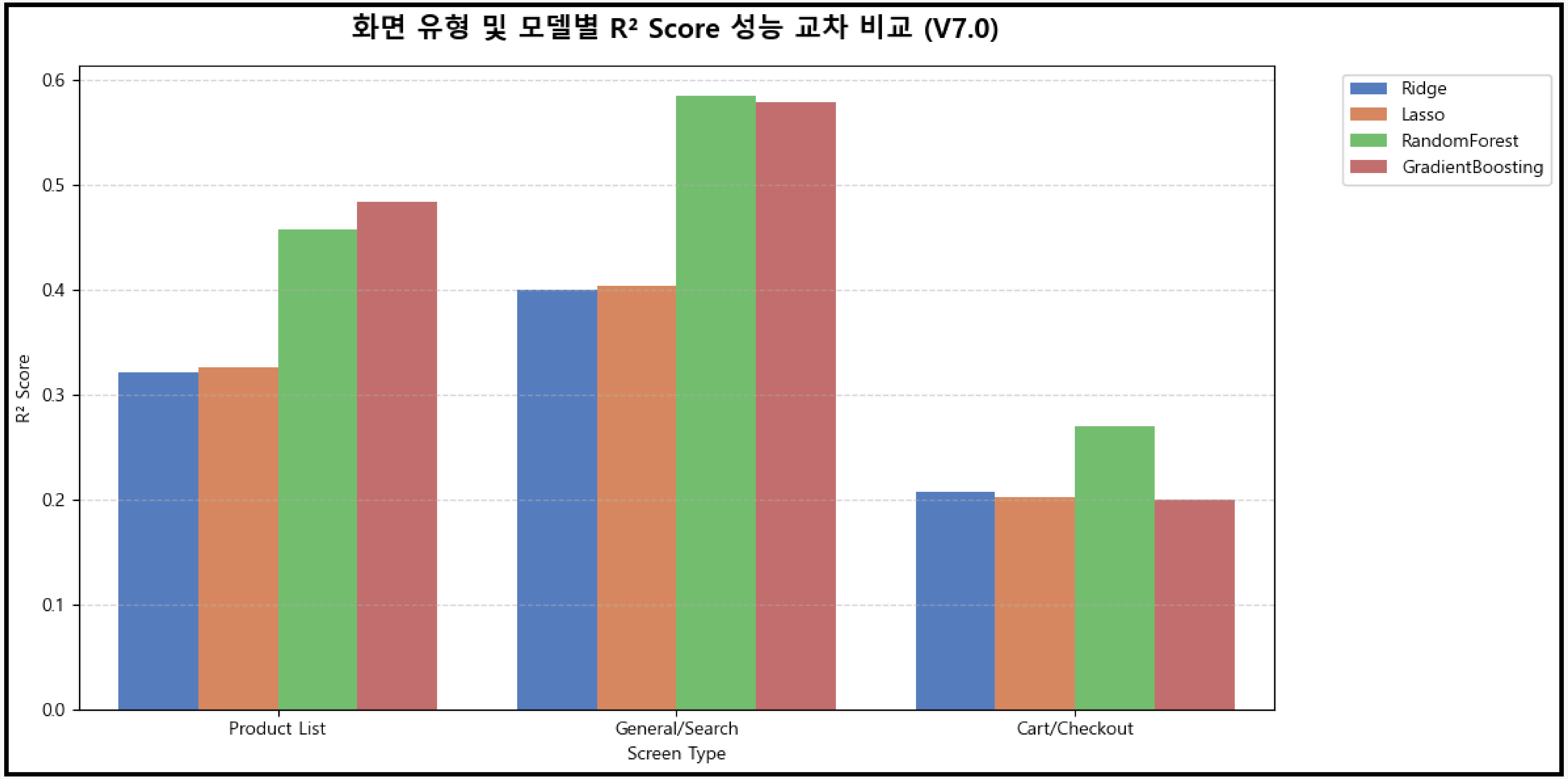


3. Cart/Checkout
   - Top Complexity Inducers
     - X_equilibrium_score : +1.6160
     - X_local_density_max : +0.6814
     - X_proportion_score : +0.6022
   - Top Complexity Reducers
     - X_input_density : -6.3261
     - X_scrollable_density: -3.8259
     - X_size_consistency : -0.2781

Results


화면 유형별 인지 부하(UI 복잡도) 구성 차원 비율 비교 (V8.0)
Product List
Global (All)
General/Search
Cart/Checkout
화면 유형
0.0
0.2
0.4
0.6
0.8
1.0
Weight Ratio (정규화된 가중치 합)
Cognitive Dimensions
구조적 깊이
균형 및 대칭
레이아웃 정렬
상호작용 부담
쇼핑 의미
시각적 복잡도
요소 밀도
텍스트 부담

Results


화면 유형 및 모델별 R² Score 성능 교차 비교 (V7.0)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
R² Score
Product List
General/Search
Cart/Checkout
Screen Type
Ridge
Lasso
RandomForest
GradientBoosting

Results

Product List

- Endless Grid Congestion
- Card-level Overload
- Structural Displacement

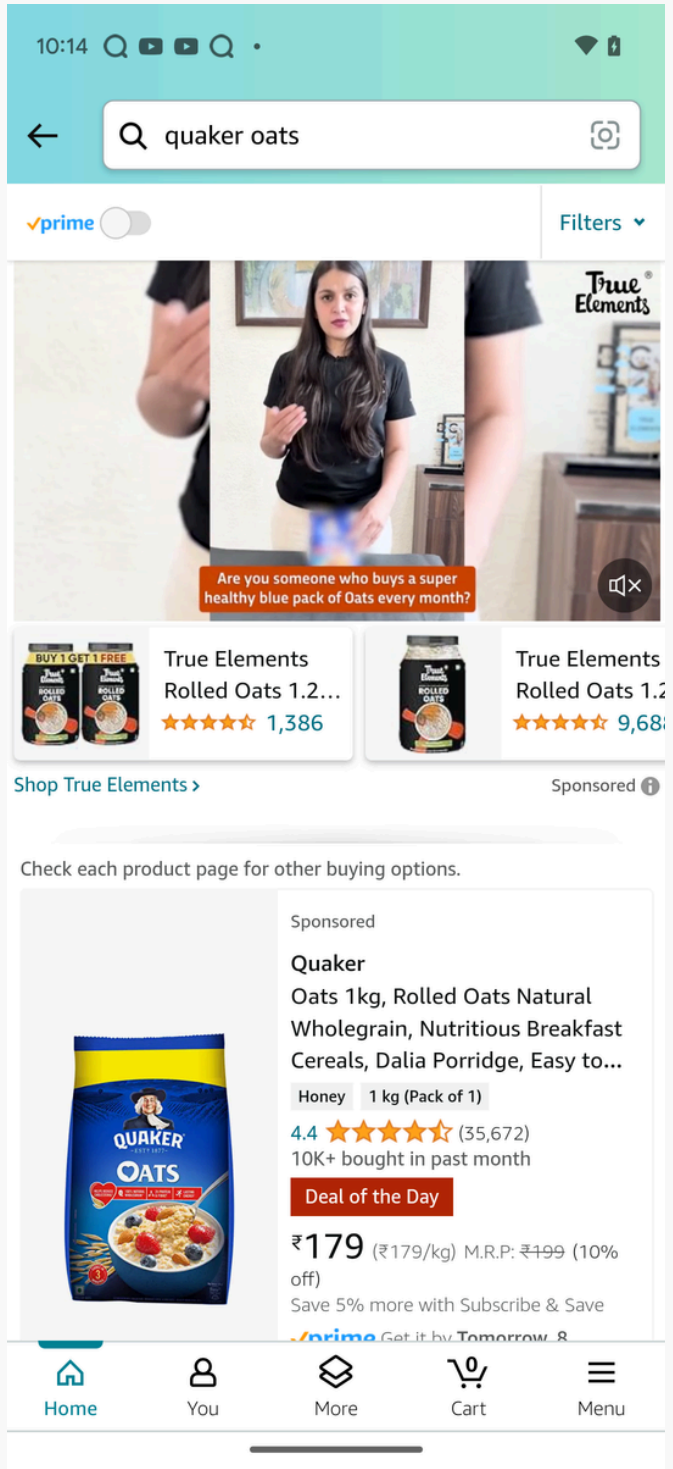
General/Search

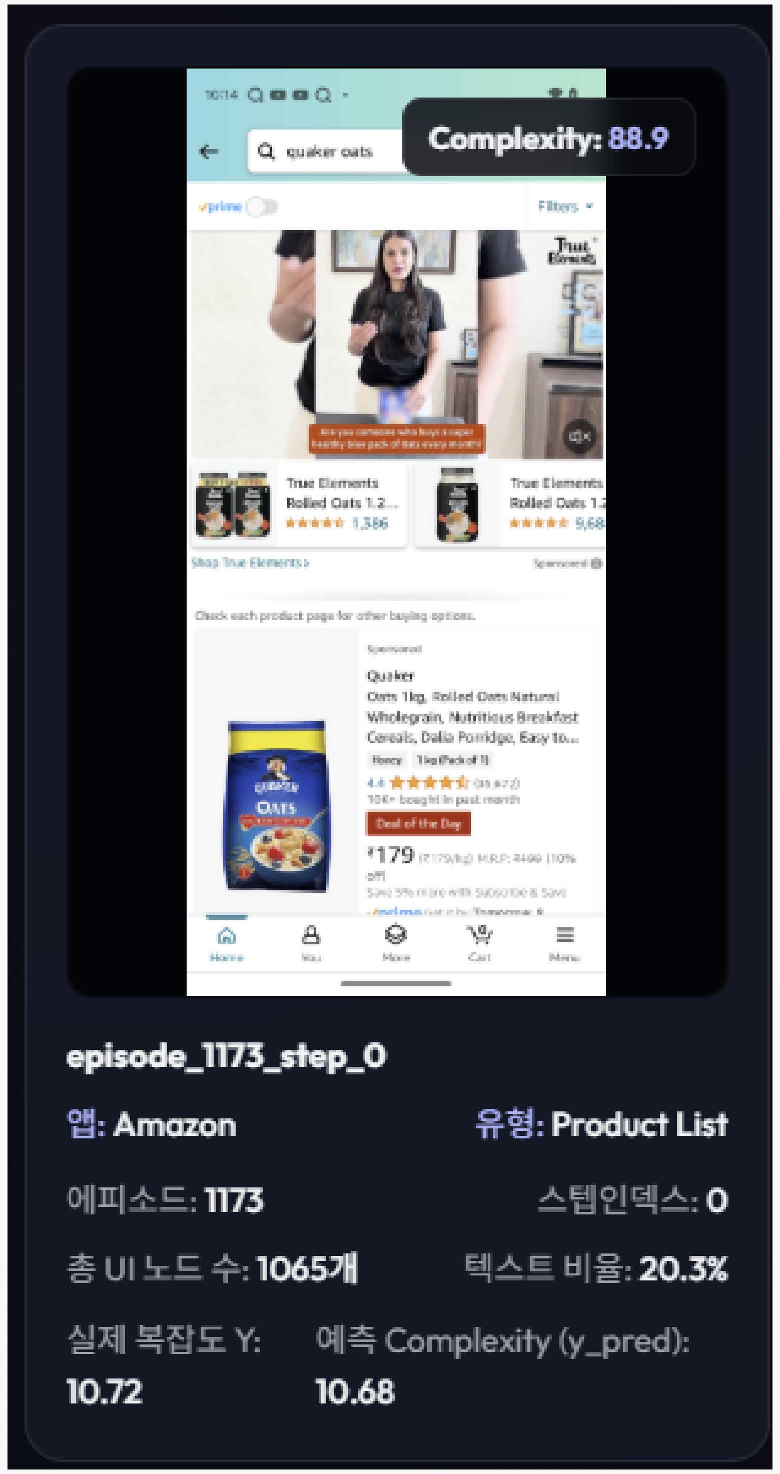
- Interactive Overload
- num_scrollable
- Anchored Simplification

Cart/Checkout

- Flat Hierarchy
- Transactional Chunks
- Input, Simplification

Results





High Complexity

Endless Grid Congestion (scrollable_density: 1.0)

- 100% scrollable area with zero visual resting space.

Flat Hierarchy (equilibrium_score: 0.98)

- Overly symmetrical layout lacking clear visual focal points.

Card-Level Overload (local_density_max: 1.0)

- Extreme concentration of elements inside a single card.

BLACK BOX Model

Modeling

Model	Input	Architecture	Objective
Image-only	Screen shot image	CNN→ FC Layer → Y	Predict complexity using only image pixels
Metadata-only	Accessibility tree features	MLP → Y	Predict complexity using only structured features
Fusion	Image + Metadata	CNN+ MLP → Concat → FC Layer → Y	Combine image and structural information

Modeling – Image only

Approach: Raw Pixels (Edges/Gradients) over Handcrafted Features

Input

224x224 RGB Data

Model

CNN
+ Regression Head

Bottleneck: Black-box (No Structural Explicability)

Modeling – Metadata only

Approach: Non-linear Feature Interaction over Linear Constraints

Input

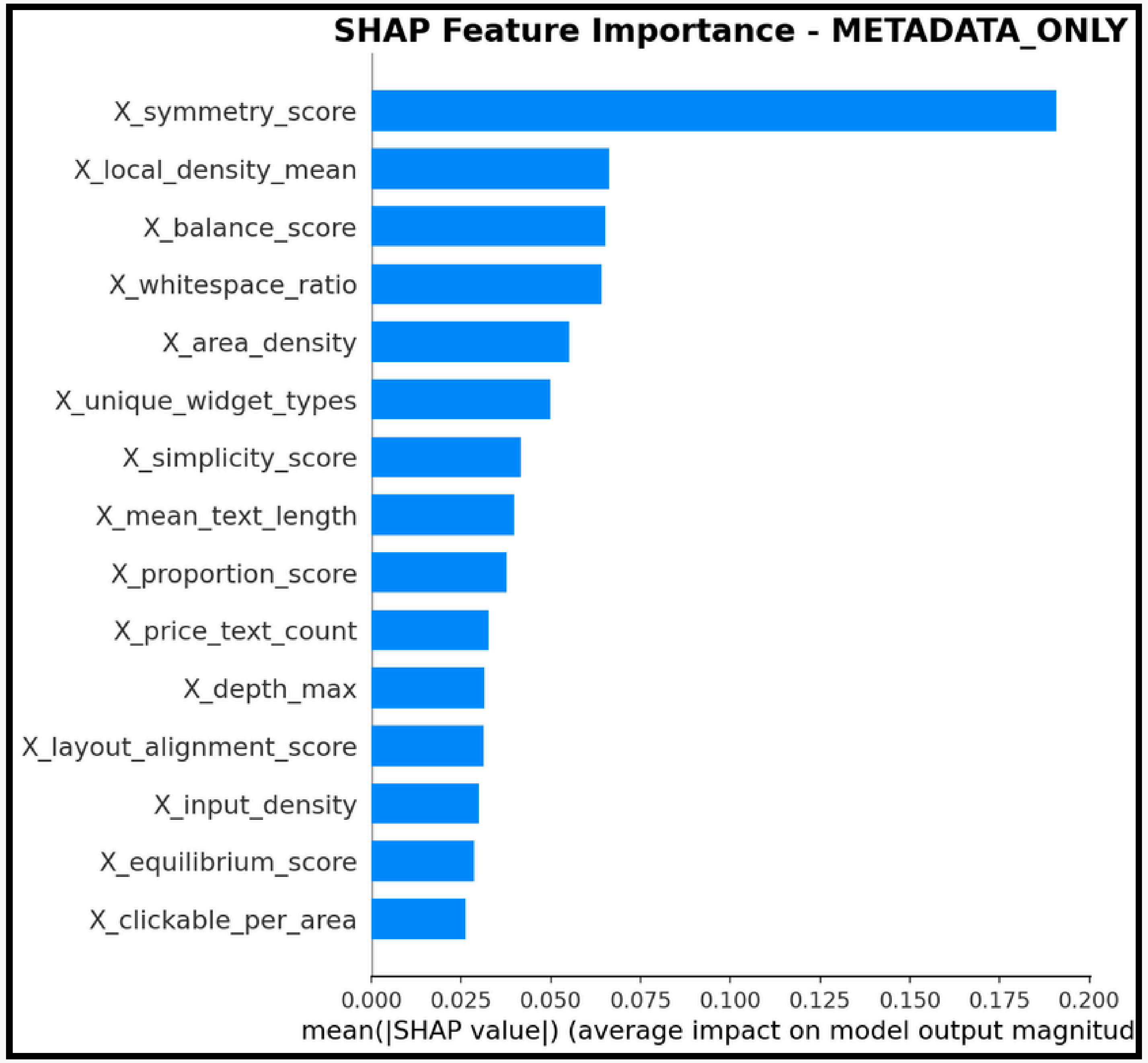
28 UI Metadata Features

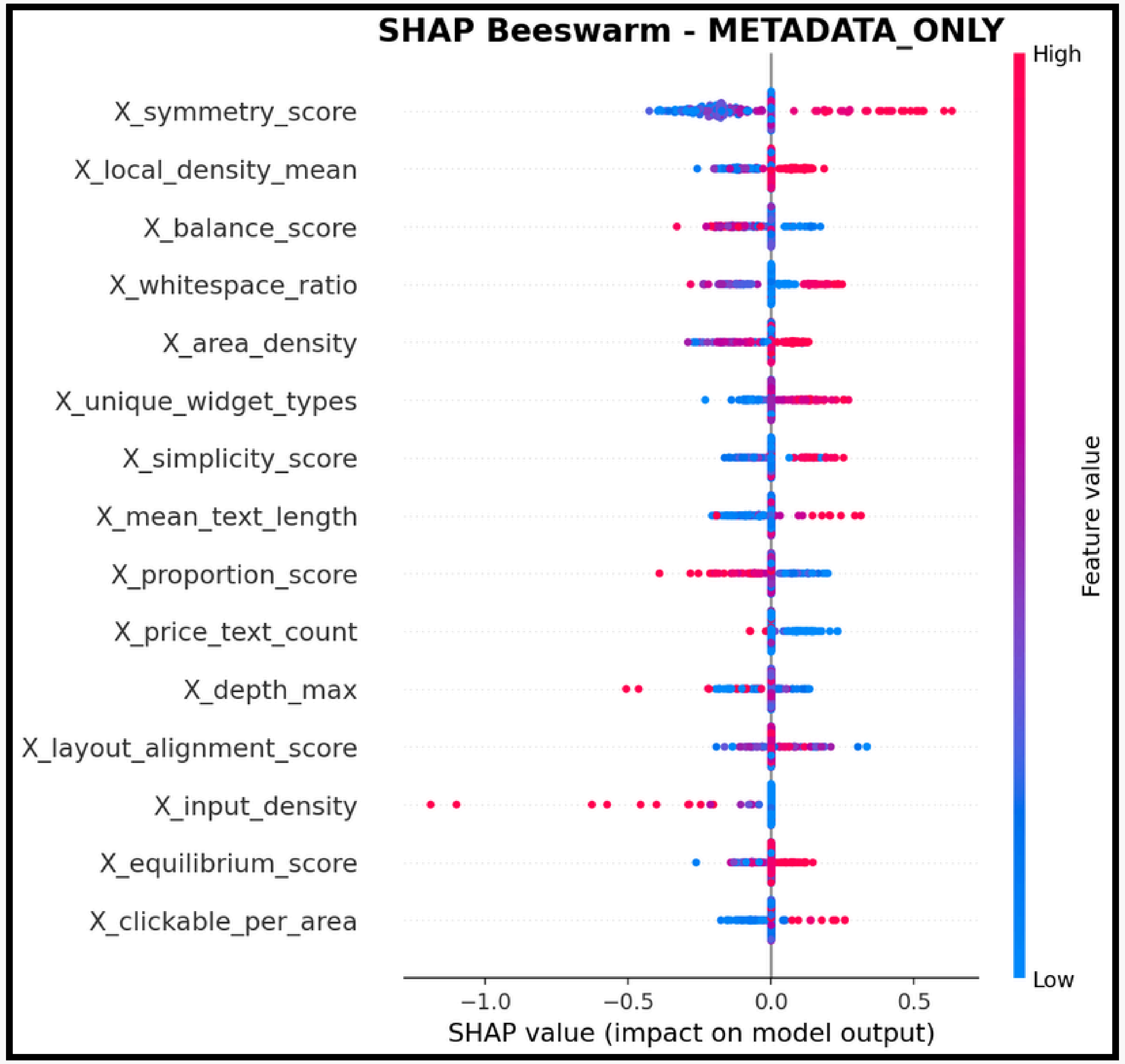
Model

Fully-Connected MLP
+ BatchNorm
+ ReLU
+ Dropout (0.3)

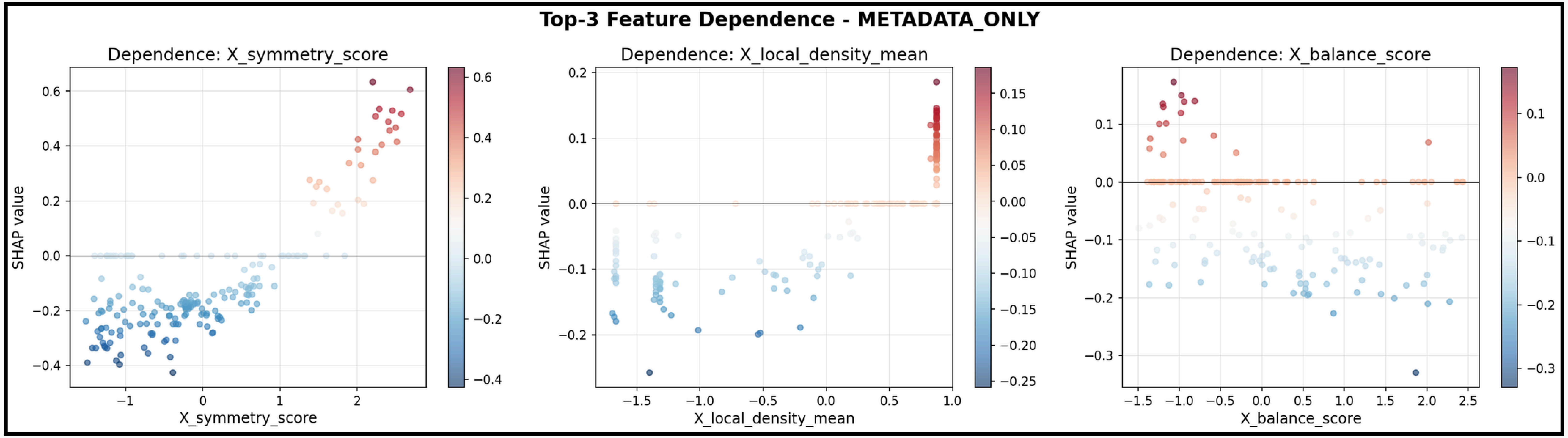
Bottleneck: Severe Overfitting
(Excessive metadata noise & zero visual signals)

Modeling – Metadata only (SHAP)





Model Comparison



Modeling – Fusion

Approach: Late-Fusion Multimodal Integration (Visual CNN + Structural MLP)

Model

Input

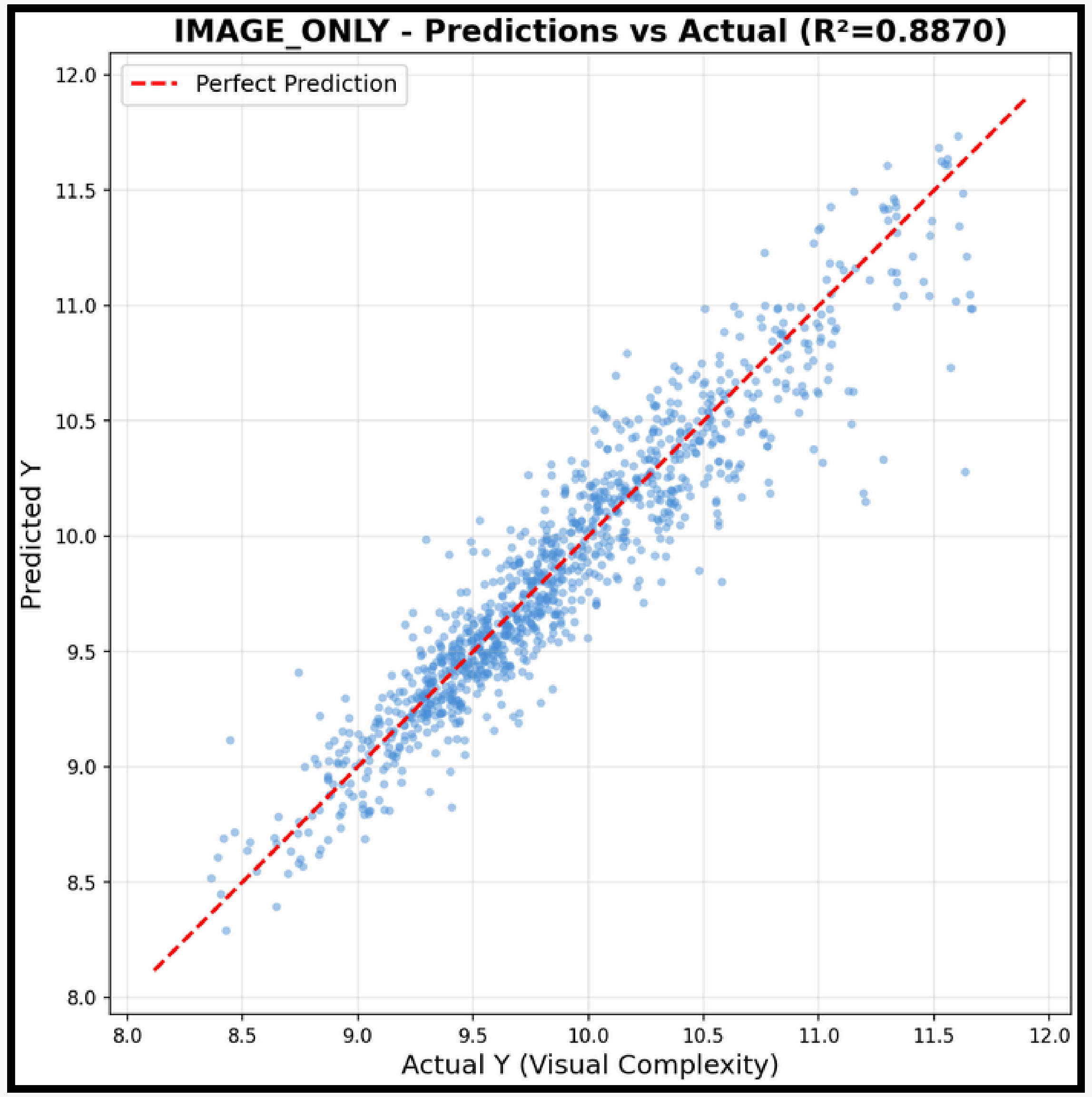
Multimodal Data

Late-Fusion Network

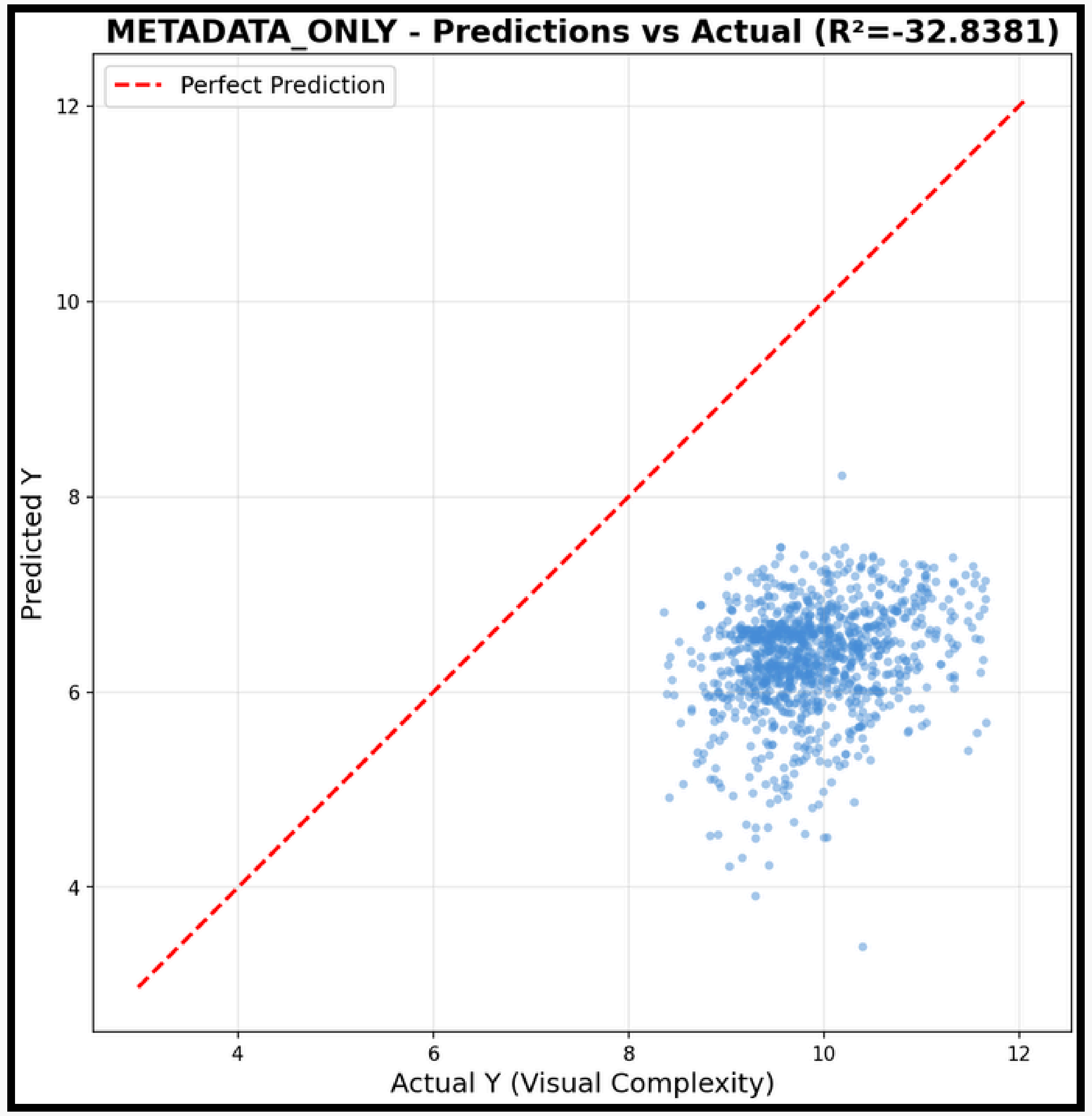
Bottleneck: Regularization Bottleneck
(Noisy metadata features degrading pristine image signals)

Model Comparison

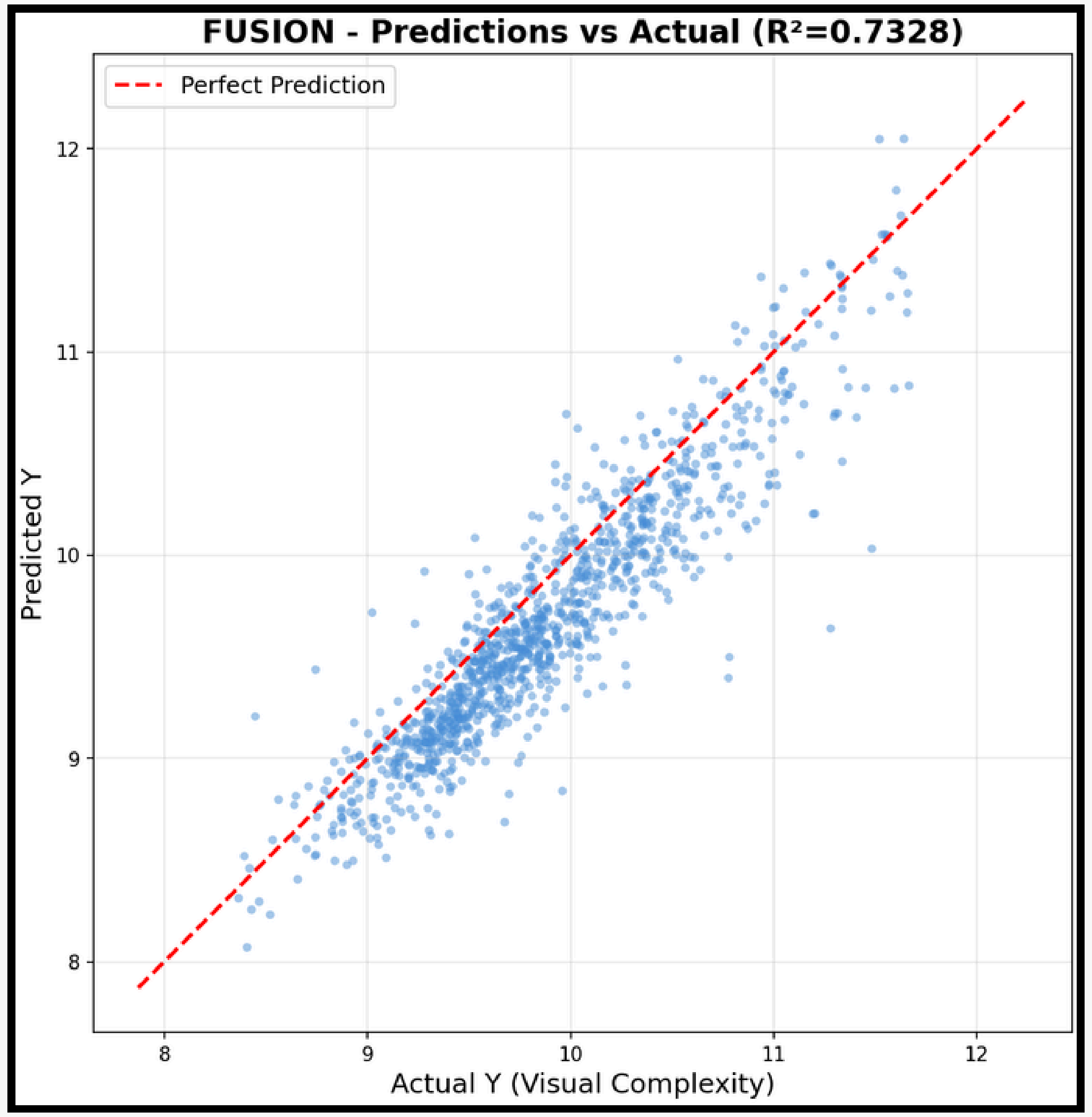
Image-only



Metadata-only



Fusion



Fusion

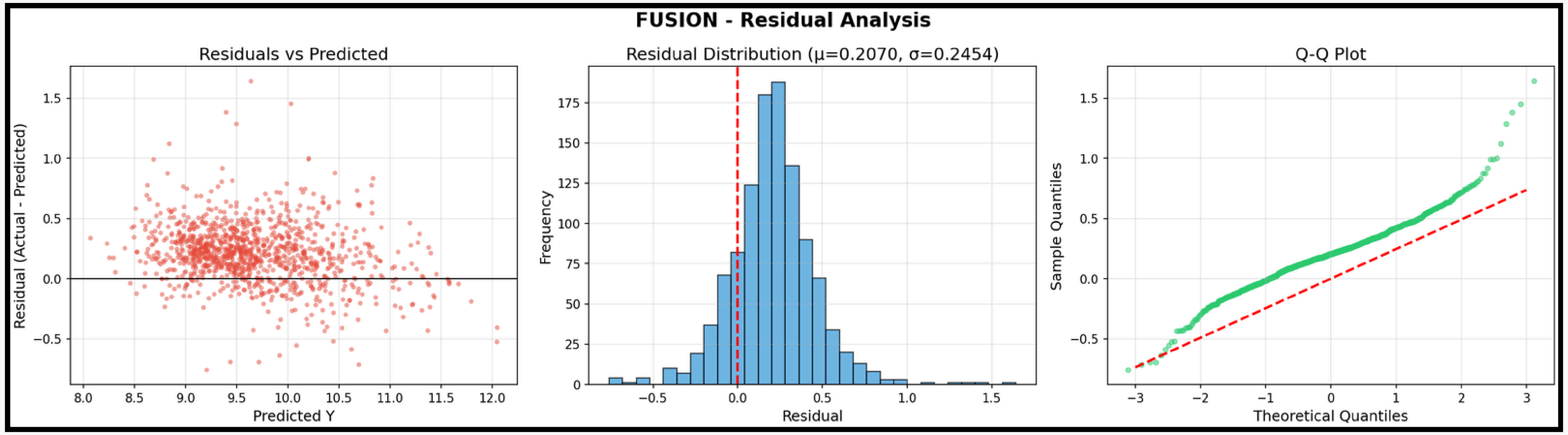
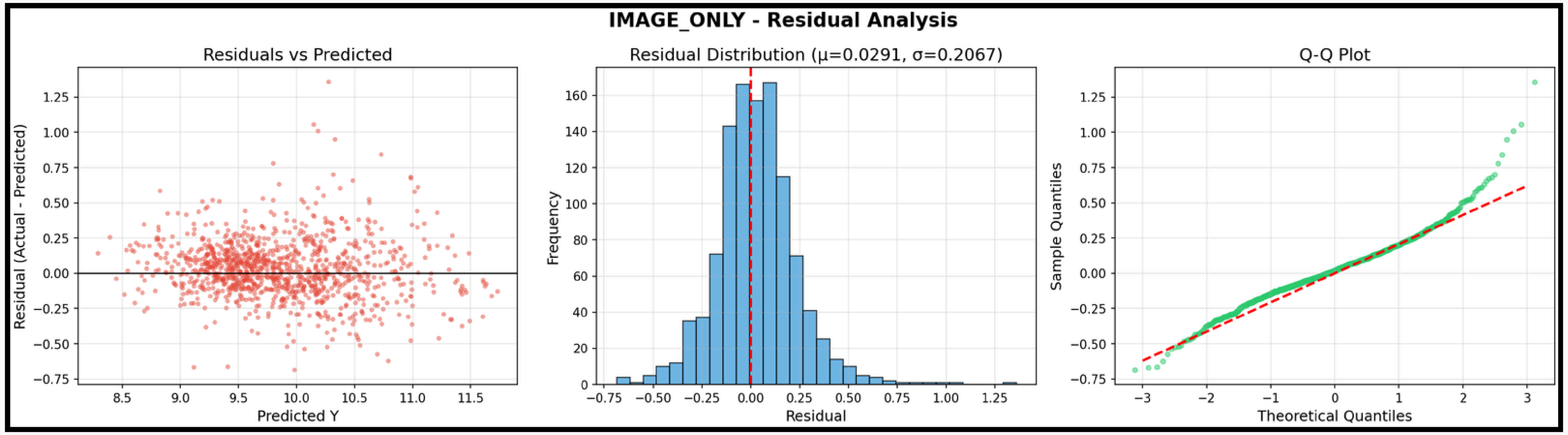
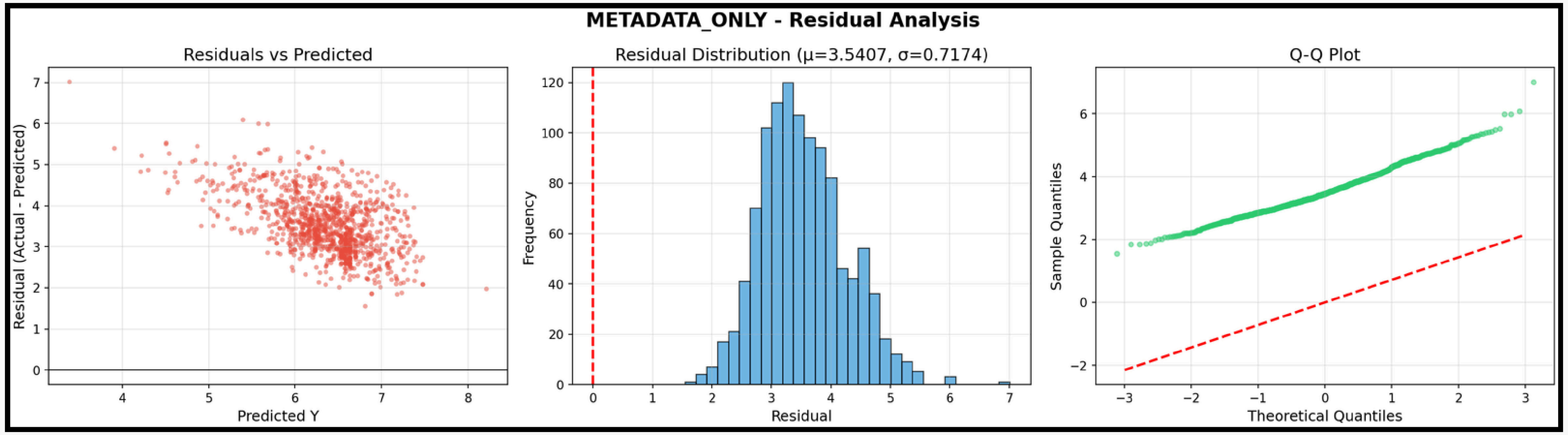


Image-only



Metadata-only



Model Comparison

Model	R^2	MSE
Fusion	0.7818	0.0772
Image-only	0.8901	0.0389
Metadata-only	-28.3771	10.395

Model Comparison

Image-only

Best Performance

➔ Proves pixel data (Edges/Gradients) is the core driver of complexity

Metadata-only

Severe Overfitting

➔ High structural noise & absence of visual signals

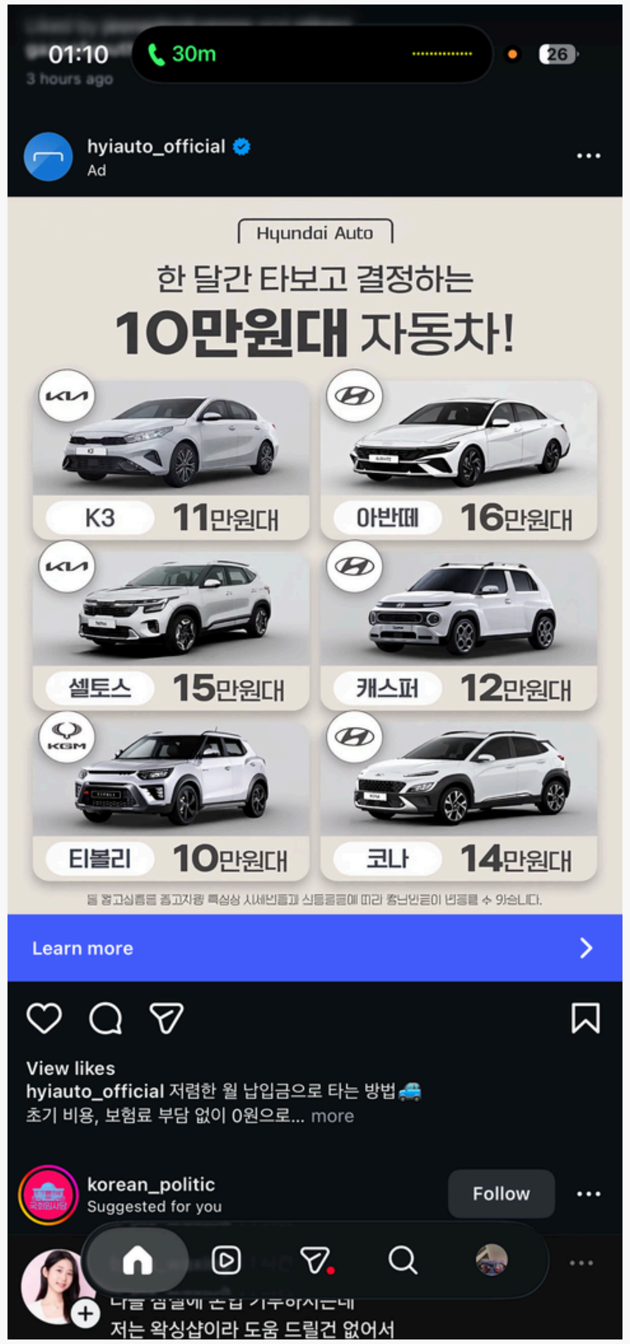
Fusion

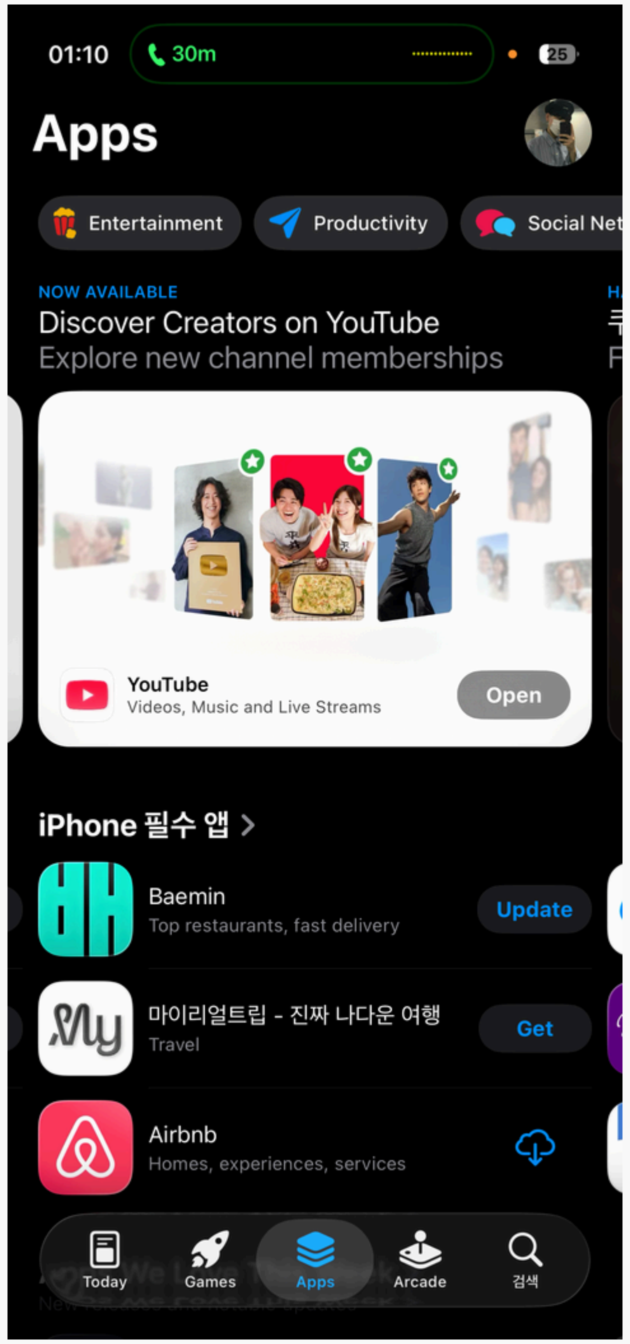
Performance Drop

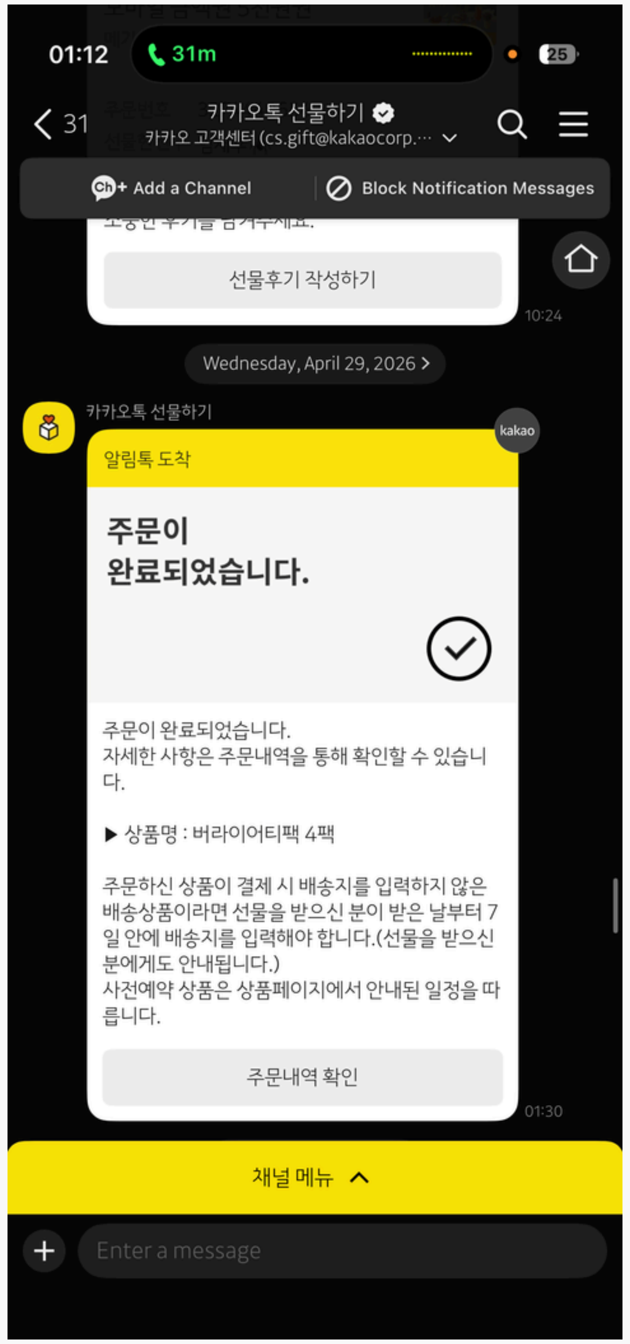
➔ Noise coupling triggers a Regularization Bottleneck (R^2=0.73)

Case Study

- Instagram
- AppStore
- KakaoTalk

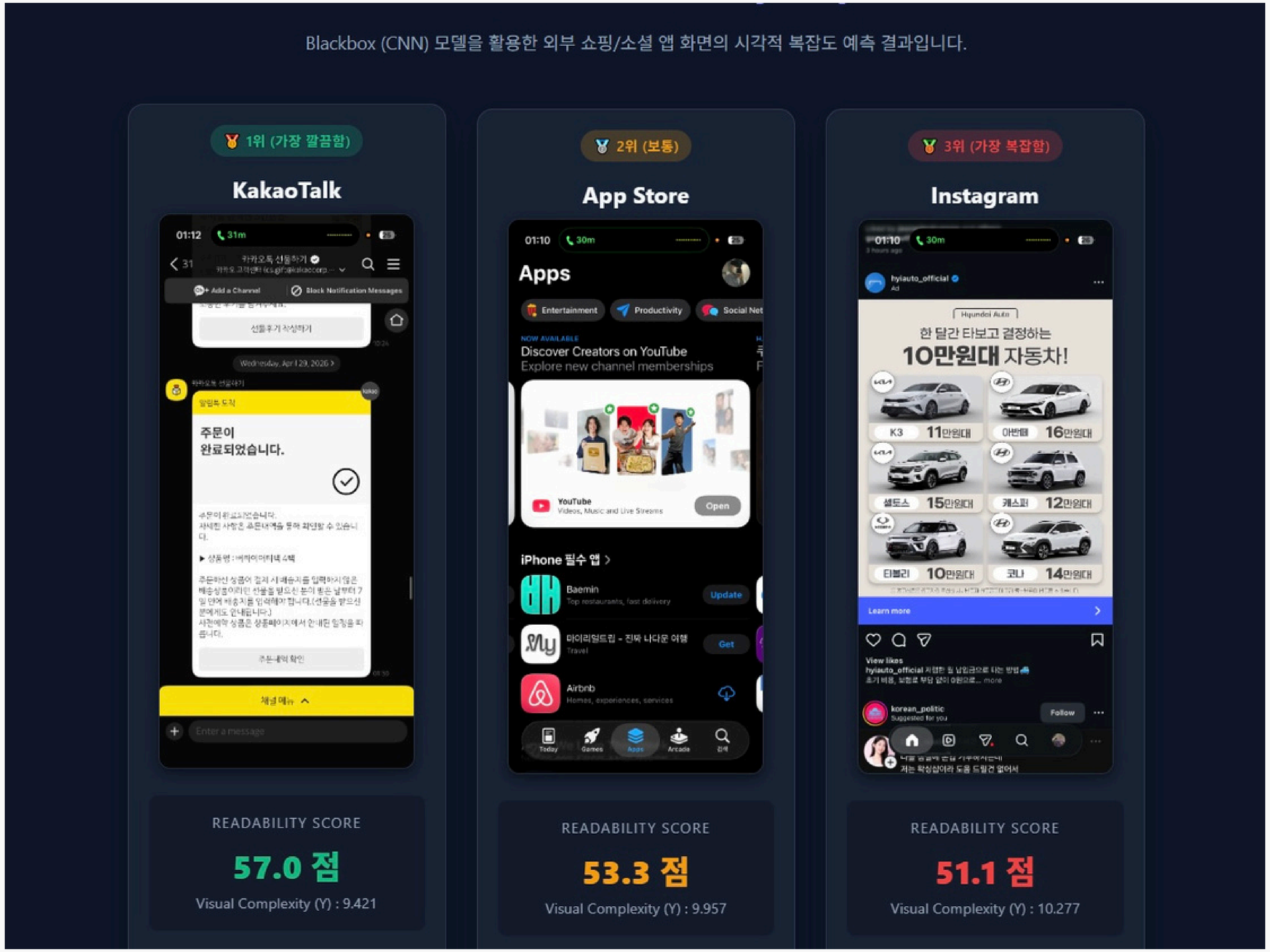






Case Study

- Instagram: 51.1
  (Y = 10.277)
- AppStore: 53.3
  (Y = 9.957)
- KakaoTalk: 57.0
  (Y = 9.421)



Implications

Existing Studies

- Uniform Evaluation
- Ignored User Patterns
- Limited Mobile Focus

This Research

- Pre-classification
- Tailored Troubleshooting
- Feature-Driven Formulation

Limitation

- Absence of Direct 'UI Readability Score' Labels
- Although an image-only model eliminates the dependency on JSON metadata, its lack of explainability remains a significant limitation.

Future Research Directions

Flow 1. Model Lightweighting & Universal Applicability

Flow 2. Expansion into Multi-Device Environments

Flow 3. Business Advanced Optimization & Personalized UI Recommendations

Contribution Proportion

성명	MEMBER A	MEMBER B	MEMBER C	MINHA KANG	MEMBER D
역할	Data Preprocessing Model - BlackBox	PPT & Research	Data Preprocessing Model - WhteBox	Presentation Topic select Preceding Research	Preceding Research Report
기여도	22.5	10	22.5	22.5	22.5